**General comments on EE2-6 Control Engineering paper 2013**

1. The students have done very well on this question, scoring approximately 80%.
   (a) This is an electric circuit modeling question and is a somewhat typical study group question.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question, asks for a physical interpretation.
      iv. A bit tricky since it uses all the results above.

   (b) This is a Nyquist diagram/Routh-Hurwitz question and is mostly typical of study group questions.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question.
      iv. Typical study group question.

2. This question combines knowledge about Nyquist analysis and the Routh-Hurwitz criterion in a slightly non-standard way for compensator design. The students did less well on this question, scoring an average mark of 64%.
   (a) Standard study group question, although part (iii) needs a little thought to deduce the phase margin.

   (b) This uses the extended Nyquist stability criterion in that it requires the determination of closed-loop stability for all possible gains. The students tend to make elementary mistakes in signs, inversions and inequalities.

   (c) Standard study group question.

   (d) Standard study group question, although they haven't done PD controllers in that context.

3. This is a root-locus type design question and is a little tricky since it involve marginally stable open-loop. The students did well, scoring approximately 70%.
   (a) Although a simple root-locus plot, some students got it wrong since the plotting rules must be interpreted correctly.

   (b) This is standard study group question, and most students did well.

   (c) Some students found the concept of PD controller a bit difficult to understand.

   (d) Some students found plotting the RL of the compensated system a little difficult, perhaps because it asked for the breakaway points.

   (e) Some students found the implementation of a controller using opamps a little difficult.